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United States Patent Application Publication

20250274047

Kind Code

A1

Publication Date

August 28, 2025

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COMMUNICATION OF FEEDBACK AND FAULT INFORMATION IN AN ISOLATED POWER CONVERTER

Abstract

A power converter includes an isolator having a first terminal, a second terminal, a third terminal, and a fourth terminal. A first circuit has a first terminal coupled to the first terminal of the isolator and has a second terminal coupled to the second terminal of the isolator. A second circuit has a first terminal coupled to the third terminal of the isolator and has a second terminal coupled to the fourth terminal of the isolator. The second circuit includes a controller configured to transmit a control signal through the isolator to the first circuit. The control signal includes a first indication to turn on power transmission through the isolator from the first circuit to the second circuit, a second indication of a status, and a third indication to turn off power transmission.

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Family ID: 1000007715566

Appl. No.: 18/587068

Filed: February 26, 2024

Publication Classification

Int. Cl.: H02M3/16 (20060101); H03K3/037 (20060101)

U.S. Cl.:

CPC H02M3/16 (20130101); H03K3/037 (20130101);

Background/Summary

BACKGROUND

[0001] An isolated power converter includes an isolation barrier, e.g., a transformer, between a first circuit (also referred to as the “primary side circuit”) and a second circuit (also referred to as the “secondary side circuit”). The first circuit may include transistors coupled to one side of the isolation barrier, e.g., to a primary winding of a transformer. The second circuit includes a rectifier coupled to the other side of the isolation barrier, e.g., to the secondary winding of the transformer. A controller turns the transistors of the first circuit on and off to convert an input voltage, VIN, to a time-varying voltage to cause energy to be transferred through the isolation barrier. The rectifier circuit converts the time-varying voltage from the isolation barrier to a direct current (DC) voltage, VOUT, and current for powering a load.

[0002] Some isolated power converters have a communication channel through the isolation barrier by which the second circuit can communicate ON and OFF indications. When the output voltage VOUT from the power converter falls below a lower threshold, the second circuit communicates an ON indication through the communication channel to cause the first circuit to turn on power transfer through the isolation barrier. When the output voltage VOUT from the power converter reaches an upper threshold, the second circuit communicates an OFF indication through the communication channel to cause the first circuit to turn off power transfer through the isolation barrier. The ON and OFF indications help the power converter to regulate the level of the output voltage.

SUMMARY

[0003] In one example, a power converter includes an isolator having a first terminal, a second terminal, a third terminal, and a fourth terminal. A first circuit has a first terminal coupled to the first terminal of the isolator and has a second terminal coupled to the second terminal of the isolator. A second circuit has a first terminal coupled to the third terminal of the isolator and has a second terminal coupled to the fourth terminal of the isolator. The second circuit includes a controller configured to transmit a control signal through the isolator to the first circuit. The control signal includes a first indication to turn on power transmission through the isolator from the first circuit to the second circuit, a second indication of a status, and a third indication to turn off power transmission.

[0004] In another example, a power converter includes an isolator having a first terminal, a second terminal, a third terminal, and a fourth terminal. A first circuit has a first terminal coupled to the first terminal of the isolator and has a second terminal coupled to the second terminal of the isolator. A second circuit has a first terminal coupled to the third terminal of the isolator and has a second terminal coupled to the fourth terminal of the isolator. The first circuit includes a decoder configured to decode a control signal received from the second circuit through the isolator. The control signal includes a first indication to turn on power transmission through the isolator to the second circuit, a second indication of a status of the second circuit, and a third indication to turn off power transmission.

[0005] In yet another example, a power converter includes an isolator having a first terminal, a second terminal, a third terminal, and a fourth terminal. A first circuit has a first terminal coupled to the first terminal of the isolator and has a second terminal coupled to the second terminal of the isolator. A second circuit has a first terminal coupled to the third terminal of the isolator and has a second terminal coupled to the fourth terminal of the isolator. The second circuit includes a controller configured to transmit a control signal through the isolator to the first circuit. The control signal includes a first indication to turn on power transmission through the isolator from the first circuit to the second circuit, a second indication of whether the second circuit is receiving power from the isolator, and a third indication to turn off power transmission. The second indication is indicated in a length of a signal pulse. The first circuit receives the control signal from the second circuit through the isolator and determines the second indication.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a block diagram of an isolated power converter, in an example.

[0007] FIG. 2 is a block diagram of the isolated power converter, in another example.

[0008] FIG. 3 is a timing diagram illustrating the operation of the isolated power converter of FIG. 2, in an example.

[0009] FIG. 4 is a circuit diagram of a controller usable in the isolated power converter of FIG. 2, in an example.

[0010] FIG. 5 is a timing diagram illustrating the operation of the controller of FIG. 4, in an example.

[0011] FIG. 6 is a circuit diagram of a decoder usable in the isolated power converter of FIG. 2, in an example.

[0012] FIG. 7 is a timing diagram illustrating the operation of the decoder of FIG. 6, in an example.

[0013] FIG. 8 is a block diagram of the isolated power converter in another example in which a multi-bit fault indication can be communicated through an isolation barrier.

[0014] FIG. 9 is a circuit diagram of a controller usable in the isolated power converter of FIG. 8, in an example.

[0015] FIG. 10 is a circuit diagram of a decoder usable in the isolated power converter of FIG. 8, in an example.

[0016] FIG. 11 is a timing diagram illustrating the generation of a multi-bit fault indication, in an example.

[0017] FIG. 12 is a circuit schematic of a switching detect circuit usable in the isolated power converters described herein, in an example.

[0018] FIG. 13 is a timing diagram illustrating the operation of the switching detect circuit of FIG. 12, in an example.

DETAILED DESCRIPTION

[0019] The same reference numbers or other reference designators are used in the drawings to designate the same or similar (either by function and/or structure) features.

[0020] FIG. 1 is a block diagram of an isolated power converter **100**, which includes a first circuit **110**, an isolator **120**, a second circuit **130**, and capacitors C1 and C2. Isolated power converter **100** converts an input voltage VIN into an output voltage VOUT. In the example of FIG. 1, first and second circuits **110** and **130** and isolator **120** are provided in the same integrated circuit (IC) package **101**. First circuit **110** may be fabricated on one semiconductor die. Second circuit **130** may be fabricated on another semiconductor die. Isolator **120** may be fabricated within the lead frame to which either or both of the dies containing first and second circuits **110** and **130** are attached. The dies and isolator **120** may be packaged together in the same IC package **101**. Capacitors C1 and C2 may be external but coupled to the IC package **101**. In another example, capacitors C1 and C2 may be included as well on IC package **101**.

[0021] First circuit **110** has terminals **111**, **112**, **113**, **114**, **115**, and **116**. Capacitor C1 is coupled across terminals **115** and **116**. First circuit **110** includes a decoder **118**. Isolator **120** includes coils L1, L2, L3, and L4. Second circuit **130** has terminals **131**, **132**, **133**, **134**, **135**, and **136**. Second circuit **130** includes a controller **138**.

[0022] In the example of FIG. 1, isolator **120** includes coils L1, L2, L3, and L4. Coils L1 and L2 may form a first transformer, and coils L3 and L4 may form a second transformer. The first and second transformers may share a core or may have separate cores. In other examples, isolator **120** may be other than coils forming a transformer. Isolator has terminals **121**, **122**, **123**, **124**, **125**, **126**, **127**, and **128**.

[0023] Coil L1 is coupled to terminals **121** and **122**, and terminals **121** and **122** are coupled to

terminals **111** and **112**, respectively, of first circuit **110**. Coil L2 is coupled to terminals **125** and **126**, and terminals **125** and **126** are coupled to terminals **131** and **132**, respectively, of second circuit **130**. Coil L3 is coupled to terminals **123** and **124**, and terminals **123** and **124** are coupled to terminals **113** and **114**, respectively, of first circuit **110**. Coil L4 is coupled to terminals **127** and **128**, and terminals **127** and **128** are coupled to terminals **133** and **134**, respectively, of second circuit **130**.

[0024] Power is transferred from first circuit **110** through coils L1 and L2 to second circuit **130**. Data is transferred from second circuit **130** through coils L3 and L4 to first circuit **110**. A data channel including coils L4 and L3 of isolator **120** may be used by second circuit **130** to communicate with first circuit **110**, for example by transmitting TURN-ON and TURN-OFF indications as well as status and fault indications, as described below. Second circuit **130** generates the TURN-ON indication in response to the output voltage VOUT falling below a lower threshold. The TURN-ON indication is received from isolator **120** and decoded by decoder **118** of first circuit **110**. First circuit **110** responds to the TURN-ON indication by turning on power transfer through coils L1 and L2 to second circuit **130**. First circuit **110** may include switching devices, e.g., transistors, to convert input voltage VIN to a switching waveform to be applied across coil L1. Second circuit **130** may include a rectifier to convert the received switching waveform from coil L2 to a direct current (DC) voltage to charge capacitor C2 thereby increasing the output voltage VOUT.

[0025] In response to the output voltage VOUT reaching an upper threshold, second circuit **130** generates the TURN-OFF indication, which is transmitted through coils L3 and L4 to first circuit **110**. First circuit **110** receives and decodes, e.g., by decoder **118**, the OFF indication and, in response, turns off power transfer through coils L1 and L2. As a result of the cessation of power transfer, the output voltage VOUT decreases, and the process repeats. Accordingly, second circuit **130** generates the TURN-ON and TURN-OFF indications as feedback signals to first circuit **110** to thereby regulate the level of the output voltage VOUT.

[0026] FIG. 2 is a block diagram of isolated power converter **100** including additional components within first circuit **110** and second circuit **130**, in an example. First circuit **110** includes decoder **118**, a demodulator **202**, control and fault management **208**, and a power stage **206**. Demodulator **202** includes terminals **237a** and **237b** which are coupled through terminals **113** and **114** of first circuit **110** to terminals **133** and **134** of coil L3. An output terminal **237c** of demodulator **202** is coupled to an input **118a** of decoder **118** and provides an RX_OUT_PLS signal **237** to decoder **118**. Decoder **118** has output terminals **118b** and **118c** which are coupled to inputs **208a** and **208b**, respectively, of control and fault management **208**. Decoder **118** provides a PS_ON_PRI signal **239** and a FAULT_STATE **245** to control and fault management **208**. Control and fault management **208** has an output terminal **208c** which is coupled to an input terminal **206a** of power stage **206**. Power stage **206** has terminals **206b** and **206c**, which are coupled through first circuit terminals **111** and **112** to terminals **121** and **122** of coil L1. Capacitor C1 has a first terminal coupled through terminal **115** to power stage **206**. Capacitor C1 has a second terminal coupled through terminal **116** to a ground terminal **206d** of power stage **206**.

[0027] Second circuit **130** includes controller **138**, a modulator **222**, a rectifier **224**, and a comparator **230**. Modulator **222** has terminals **222a** and **222b** coupled through terminals **133** and **134** of second circuit **130** to terminals **127** and **128** of coil L4. Rectifier **224** has terminals **224a** and **224b** coupled through terminals **131** and **132** of second circuit **130** to terminals **125** and **126** of coil L2. Rectifier **224** has an output terminal **224c** coupled to an input terminal **138a** of controller **138**. Rectifier **224** has a switching detect circuit **226**, which is coupled to output terminal **224c**. An example of a switching detect circuit **226** is provided in FIG. 12, described below. Switching detect circuit **226** generates a signal SEC_SW_DET **235** to controller **138**. Comparator **230** has a positive (+) input terminal, a negative (−) input terminal, and output terminal **230a**. In this example, the positive input terminal receives a reference voltage (REF), and the negative terminal is coupled to a

terminal of capacitor C2 and accordingly receives the output voltage VOUT. The other terminal of capacitor C2 is coupled to terminal 136 of second circuit 130. Terminal 136 of second circuit 130 is coupled to a ground terminal 224d of rectifier 224. The output terminal 230a is coupled to an input 138b of controller 138. Comparator 230 generates a signal PS_ON_SEC 231 to controller 138. Controller 138 has an output terminal 138c, which is coupled to an input terminal 222c of modulator 222. Controller 138 generates signal TX_ON_PLS 233 to modulator 222.

[0028] The operation of isolated power converter 100 is described with respect to the timing diagram of FIG. 3. When power transfer through coils L1 and L2 is off, capacitor C2 begins to discharge into the load (not shown) powered by isolated power converter 100. Accordingly, output voltage VOUT decreases. Comparator 230 compares the magnitude of output voltage VOUT to the reference voltage REF. Comparator 230 may implement a hysteresis voltage difference (HYST) centered on the reference voltage REF (e.g., $REF+HYST/2$ and $REF-HYST/2$). When the voltage VOUT falls below $REF-(HYST/2)$, comparator 230 forces signal PS_ON_SEC 231 to a logic high state, as indicated by rising edge 302 in FIG. 3. Controller 138 responds to a logic high assertion of signal PS_ON_SEC 231 by generating the TX_ON_PLS signal 233 to provide a TURN-ON indication 304 to modulator 222.

[0029] In one example, the TURN-ON indication 304 includes signaling of the TX_ON_PLS signal 233 that is characterized by the TX_ON_PLS signal 233 including a pair of pulses 304a and 304b with pulse 304b being longer than pulse 304a. In a specific example, the TURN-ON indication 304 is characterized by: (a) the TX_ON_PLS signal 233 including a pair of pulses 304a and 304b, (b) pulse 304a having a pulse width 332 of 25 ns, (c) pulse 304b having a pulse width 334 of at least 50 ns, and (d) the time period 336 between pulses 304a and 304b being 30 ns. In other examples, the aforementioned time periods can have different values.

[0030] In response to receipt of the TURN-ON indication 304, modulator 222 modulates a carrier signal using the TX_ON_PLS signal 233, which includes the TURN-ON indication. The TX_ON_PLS signal-modulated carrier signal is transferred through coils L4 and L3 of isolator 120 to demodulator 202. Demodulator 202 demodulates the carrier signal to recover the TX_ON_PLS signal 233 as RX_OUT_PLS signal 237. The RX_OUT_PLS signal 237 includes the TURN-ON indication, indicated in FIG. 3 as TURN-ON indication 338. TURN-ON indication 338 includes pulses 338a and 338b separated by a time period 339. The widths of pulses 338a and 338b and the time period 339 generally match the widths of corresponding pulses 304a and 304b and time period 336. Decoder 118 receives the RX_OUT_PLS signal 237, which includes the TURN-ON indication 338 and, in response, asserts (e.g., logic high at rising edge 345) the PS_ON_PRI signal 239 to control and fault management 208. Control and fault management 208 responds to the logic high assertion of the PS_ON_PRI signal 239 by causing power stage 206 to turn on and begin transferring power from coil L1 to coil L2 of isolator 120 as indicated at 317. Turning on power stage 206, for example, causes a time-varying current to occur between terminals 121 and 122 of coil L1. Rectifier 324 rectifies the received power from a time-varying current to a DC current to charge capacitor C2 thereby causing the output voltage VOUT to increase.

[0031] When the output voltage VOUT exceeds $REF-(HYST/2)$, comparator 230 forces the PS_ON_SEC signal 231 to a logic low state at falling edge 352. Controller 138 responds to a logic low assertion of the PS_ON_SEC signal 231 by generating the TX_ON_PLS signal 233 to provide a TURN-OFF indication 354 to modulator 222. In one example, the TURN-OFF indication 354 includes a single pulse 356 of a prescribed pulse width 358 of, for example, at least 50 ns.

[0032] Modulator 222 communicates the TX_ON_PLS signal 233 with the TURN-OFF indication 354 through the isolator 120 as explained above. Demodulator 202 recovers the TX_ON_PLS signal 233 as the RX_OUT_PLS signal 237 having the TURN-OFF indication (TURN-OFF indication 360). Decoder 118 responds to the TURN-OFF indication by forcing the PS_ON_PRI signal 239 to a logic low state at falling edge 362. Control and fault management 208 responds to the logic low state of the PS_ON_PRI signal 239 by turning off power stage 206 thereby turning

off power transfer between coils L1 and L2. The time-varying currents through coils L1 and L2 then cease as indicated at **365**. As a result of the cessation of power transfer through isolator **120**, capacitor C2 begins to discharge into the load and output voltage VOUT begins to fall. The process repeats.

[0033] In one example, controller **138** can generate the TX_ON_PLS signal **233** to provide a status indication between the TURN-ON and TURN-OFF indications **304** and **354**. In the example of FIG. 3, the status indication is encoded as a status bit in the TX_ON_PLS signal **233** towards the end of the TURN-ON indication **304**. Switching detect circuit **226** detects whether power is being transferred to second circuit **120** from first circuit **110** through isolator **120** and asserts the SEC_SW_DET signal **235** to a first logic state if power is being transferred through isolator **120** or to a second logic state if power is not being transferred through isolator **120**. In the example of FIG. 3, switching detect circuit **226** asserts the SEC_SW_DET signal **235** to a logic high state at rising edge **357** upon detection of power transfer through isolator **120**. In response to receipt of the SEC_SW_DET signal **235** at the logic high state, controller **138** determines that second circuit **130** is receiving power and, in response, causes the TX_ON_PLS signal **233** to transition from a logic high state to a logic low state at falling edge **349** thereby terminating pulse **304b**. Otherwise, if controller **138** does not receive a logic high assertion of the SEC_SW_DET signal **235**, indicative of the condition that power is not being transferred through isolator **120** to second circuit **130**, controller **138** causes the logic high state to logic low state transition of TX_ON_PLS **233** at a later point in time-falling edge **353**. Accordingly, whether the TX_ON_PLS signal **233**, and the corresponding RX_OUT_PLS signal **237**, is either logic low or logic high during time period **359** indicates a status of second circuit **130**, e.g., whether second circuit **130** is or is not receiving power transferred through isolator **120**.

[0034] Decoder **118** determines whether the RX_OUT_PLS signal **237** is logic low or logic high during time period **359**. A logic low (e.g., a “0”) indicates a status that second circuit **130** is receiving power transferred through isolator **120**. A logic high (e.g., a “1”) indicates a status that second circuit **130** is not receiving power from isolator **120**. Decoder **118** generates the FAULT_STATE signal **245** to indicate the corresponding 1 or 0. In this example, the FAULT_STATE signal **245** is a single bit signal. In response to receipt of the FAULT_STATE signal **245** indicating that the second circuit **130** is not receiving power through isolator **120**, control and fault management **208** may respond by, for example, discontinuing future activations of power stage **206**.

[0035] FIG. 4 is a schematic diagram of controller **138**, in an example. In this example, controller **138** includes pulse generators (PGs) **401**, **402**, **403**, and **404**, OR gates **424** and **426**, and inverters **428** and **430**. Each PG **401-404** has a clock input terminal (CLK). PGs **401** and **403** are clocked with a rising edge on their clock input terminals, and PGs **402** and **404** are clocked with a falling edge on their clock input terminals. The clock input terminals of PGs **401** and **404** receive the PS_ON_SEC signal **231**. The output OUT terminal of PG **401** is coupled to an input terminal **424a** of OR gate **424** and the clock input terminal of PG **402**. PG **401** produces a PLS1 signal **411** at its OUT terminal. The OUT terminal of PG **402** is coupled to an input of inverter **430**. PG **401** produces a PLS2 signal **412** at its OUT terminal. The output of inverter **430** is coupled to an input terminal **422a** of AND gate **422** and produces a PLS2_INV signal **413**, which is the logical inverse of the PLS2 signal **412**. Input terminal **422b** of AND gate **422** receives the PS_ON_SEC signal **231**.

[0036] The OUT terminal of PG **403** is coupled to an input terminal **424b** of OR gate **424**. The output terminal **426c** of OR gate **424** is coupled to an input terminal **426a** of OR gate **426** and produces an ON signal **417**. The OUT terminal of PG **404** is coupled to an input terminal **426b** of OR gate **426** and produces an OFF signal **427**. OR gate **426** produces the signal TX_ON_PLS **233** at its output terminal **426c**.

[0037] When clocked, each PG **401-404** produces a pulse of a particular pulse width for its

respective output signal. In one example, PG **401** produces the PLS1 signal **411** to have a pulse of width 25 ns. PG **402** produces the PLS2 signal **412** to have a pulse of width 30 ns. PG **403** produces the PLS3 signal **414** to have a pulse of width 1 μ s. PG **404** produces the OFF signal **427** to have a pulse of width 1 μ s. Different values of the pulse widths are possible in other examples. [0038] PGs **403** and **404** have a reset (RST) terminal. The SEC_SW_DET signal **235** is provided to the RST terminal of PG **403** and, through inverter **428** to the RST terminal of PG **404**. A logic high state of a signal at the RST terminal of PG **403** and PG **404** causes the respective PG to force the signal at the PG's OUT terminal to a logic low state. For example, when the SEC_SW_DET signal **235** is logic high, PG **403** resets and forces the PLS3 signal **414** to a logic low state. Similarly, when the SEC_SW_DET signal **235** is logic low, due to inverter **428**, PG **404** resets and forces the OFF signal **427** to a logic low state. When clocked, PGs **403** and **404** produce pulses having pulse widths of, for example, 1 μ s, unless reset before the end of the 1 μ s time period.

[0039] FIG. 5 is a timing diagram illustrating the operation of controller **138**. The timing diagram includes the PS_ON_SEC signal **231**, the PLS1 signal **411**, the PLS2 signal **412**, the PLS2_INV signal **413**, the PLS3 signal **414**, the ON signal **417**, the SEC_SW_DET signal **235**, the OFF signal **427**, and the TX_ON_PLS signal **233**.

[0040] Referring to both FIGS. 4 and 5, the rising edge **302** of the PS_ON_SEC signal **231** causes PG **401** to produce a pulse **502** for the PLS1 signal **411** with a pulse width of, for example, 25 ns. Through OR gates **424** and **426**, pulse **502** results in pulse **304a** for the TX_ON_PLS signal **233**. The falling edge **504** of pulse **502** clocks PG **402** thereby causing PG **402** to produce a pulse **508** for the PLS2 signal **412** with a pulse width of, for example, 30 ns. The PLS2_INV signal **413** is generated by inverter **430** is the logical inverse of the PLS2 signal **412** and, accordingly, produces a negative pulse **510**. With the PS_ON_SEC signal **231** still being at the logic high state, the rising edge **512** of the negative pulse **510** clocks, through AND gate **422**, PG **403**. PG **403** responds by producing a pulse **514** for the PLS3 signal **414** having a pulse width of, for example, 1 μ s, unless PG **403** is reset before the expiration of the 1 μ s time period. In the example of FIG. 5, the SEC_SW_DET signal **235** transitions from a logic low state to a logic high state at rising edge **520**, thereby resetting PG **403** and terminating pulse **514** at falling edge **515** before the end of the full 1 μ s time period. Otherwise, absent the logic high state of the SEC_SW_DET signal **235**, pulse **514** would have been longer (e.g., 1 μ s from PG **403**) and have terminated at falling edge **517**. Through OR gates **424** and **426**, pulse **514** results in pulse **304b** for the TX_ON_PLS signal **233**. The combination of the shorter pulse **304a** and the longer pulse **304b** represents the TURN-ON indication **304**, described above.

[0041] The falling edge **352** of the PS_ON_SEC signal **231** clocks PG **404**, which responds by producing a pulse **355** for the OFF signal **427**. Through OR gate **426**, pulse **355** results in the TX_ON_PLS signal **233** having pulse **356**. Pulse **356** represents the TURN-OFF indication **354** described above.

[0042] FIG. 6 is a schematic diagram of decoder **118**, in an example. In this example, decoder **118** includes a counter **602**, a PG **604**, a filter/delay circuit **606**, AND gates **608**, **610**, **612**, and **614**, a set(S) reset (R) flip-flop **616**, and a fault decoder circuit **618**. Fault decoder circuit **618** includes data (D) flip-flops **622** and **628**, a delay circuit **624**, a PG **626**, and a filter/delay circuit **630**.

[0043] Counter **602** has a clock terminal **602a**, an enable (EN) terminal **602b**, an output b0 terminal, and an output b1 terminal. The RX_OUT_PLS signal **237** is provided to the clock terminal **602a**. PG **604** has a clock input that receives the RX_OUT_PLS signal **237**. The OUT terminal of PG **604** is coupled to the enable terminal **602b** of counter **602**. PG **604** generates a signal CTR_EN **605** at its OUT terminal. The output b0 terminal of counter **602** is coupled to an input terminal **608a** (e.g., an inverted input) of AND gate **608** and to an input terminal **610a** of AND gate **610**. The output b1 terminal is coupled to an input terminal **608b** of AND gate **608** and to an input terminal **610b** (e.g., an inverted input) of AND gate **610**. The output terminal **608c** of AND gate **608** is coupled to an input terminal **612a** of AND gate **612**. The output terminal **610c** of

AND gate **610** is coupled to an input terminal **614a** of AND gate **614**.

[0044] Filter/delay circuit **606** has an input terminal **606a** and an output terminal **606b**. The RX_OUT_PLS signal **237** is provided to the input terminal **606a**. Filter/delay circuit **606** generates a TDLY_FLTR signal **609** at its output terminal **606b**. The output terminal **606b** of filter/delay circuit **606** is coupled to input terminals **612b** and **614b** of AND gates **612** and **614**, respectively, and to a clock terminal of D flip-flop **622**. SR flip-flop has an S terminal, an R terminal, and a Q terminal. The output terminal **612c** of AND gate **612** is coupled to the S terminal, and the output terminal **614c** of AND gate **614** is coupled to the R terminal. AND gate **612** generates a set (SET) signal **613** at its output terminal **612c**. AND gate **614** generates a reset (RST) signal **615** at its output terminal **614c**. SR flip-flop **616** generates the PS_ON_PRI signal **239** at its Q terminal. [0045] The D terminal of D flip-flop **622** is coupled to a logic high voltage terminal. The Q terminal of D flip-flop **622** is coupled to an input terminal **624a** of delay circuit **624**. An output terminal **624b** of delay circuit **624** is coupled to a clock terminal of PG **626**. Delay circuit **624** generates a TDLY2 signal **625** at its output terminal **624b**. In response to a rising edge of the TDLY2 signal **625**, PG **626** produces a pulse having a width of, for example, 250 ns at its output terminal. The output terminal of PG **626** is coupled to a clock input of D flip-flop **628** and to an input terminal **630a** of filter/delay circuit **630**. PG **626** generates a FAULT_DECODER signal **311** at its output terminal. The RX_OUT_PLS signal **237** is provided to the D input of D flip-flop **628**. D flip-flop **628** generates the FAULT_STATE signal **245** at its Q output based on the logic state of the RX_OUT_PLS signal **237** upon each rising edge of the FAULT_DECODER signal **311**. An output terminal **630b** of filter/delay circuit **630** is coupled to the RST terminal of D flip-flop **622**. The delay implemented by filter/delay circuit **630** allows time for D flip-flop **628** to be clocked before D flip-flop **622** is reset.

[0046] The example counter **602** is two-bit counter which advances its output count value with each rising edge at its clock terminal **602a**. The least significant bit is b0 and the most significant bit is bit b1. Upon initially being enabled by a logic high state at the enable terminal **602b**, bits <b1: b0> initially have the values 00. With each successive rising edge at clock terminal **602a**, bits <b1: b0> sequence through the output count values 01, 10, and 11.

[0047] PG **604** generates a pulse on the CTR_EN signal **605** having a pulse width of, for example, 150 ns. The width of the pulse of CTR_EN signal **605** is longer than the width of the first pulse **338a** of TURN-ON indication **304** and the time period **339** between the end of the first pulse **338a** and the rising edge of the subsequent pulse **338b**. The width of the pulse of CTR_EN is long enough to allow counter **602** to count both the edges of both pulses **338a** and **338b** of the TURN-ON indication **304**. In the example in which the width of pulse **338a** is 25 ns and the width of time period **339** is 30 ns, the width of pulse the pulse of CTR_EN is at least 55 ns. In one example, the width of the pulse of CTR_EN is 150 ns.

[0048] Filter/delay circuit **606** produces the TDLY_FLTR signal **609** to be a delayed version of its input signal, RX_OUT_PLS, if a pulse of the RX_OUT_PLS signal **237** has a width of at least m ns. In one example, m is 50 ns. If the width of a pulse of the RX_OUT_PLS signal **237** is less than m ns (e.g., less than 50 ns), the filter/delay circuit **606** does not generate a delayed pulse on the TDLY_FLTR signal **609**. If the width of a pulse of the RX_OUT_PLS signal **237** is at least m ns (e.g., at least 50 ns), the filter/delay circuit **606** generates a pulse on the TDLY_FLTR signal **609** with the same width as its input signal (RX_OUT_PLS signal **237**) but delayed by m ns.

Filter/delay circuit **630** in fault decoder circuit **618** is similar. It too produces a delayed signal to the RST terminal of D flip-flop **622** based on a pulse of the FAULT_DECODER signal **311** that is at least m ns (e.g., 50 ns) wide. Delay circuit **624** produces TDLY signal **625** as a delayed version (e.g., 750 ns) of the signal from the Q terminal of the D flip-flop **622**. Each of filter/delay circuits **606** and **630** may include an RC filter coupled to a buffer.

[0049] FIG. 7 is a timing diagram illustrating the operation of decoder **118**, in an example. The timing diagram includes the RX_OUT_PLS signal **237**, the CTR_EN signal **605**, the TDLY_FLTR

signal **609**, b0 (the signal at the output b0 of counter **602**), b1 (the signal at the output b1 of counter **602**), the SET signal **613**, the RST signal **615**, the PS_ON_PRI signal **239**, the TDLY2 signal **625**, and the FAULT_DECODER signal **311**.

[0050] Referring to FIGS. **6** and **7**, rising edge **702** of the first pulse **338a** of the TURN-ON indication **304** causes counter **602** to advance its output count value thereby causing <b1:b0> to be advanced from 00 to 01. With b0 equal to logic 1 and b1 equal to logic 0, the logic state of the signal at the input **612a** of AND gate **612** is logic 1. The width of pulse **338a** is not long enough for filter/delay circuit **606** to generate a delayed output pulse and thus the input **612b** of AND gate **612** remains a logic 0. Accordingly, the SET signal **613** is logic 0, as indicated at **727**.

[0051] The next rising edge **708** of the second pulse **338b** of the RX_OUT_PLS signal **237** causes counter **602** to again increment its output count value from 01 to 10. With b1 equal to logic 1 and b0 equal to logic 0, the output signal from AND gate **608** begins logic 1. The width of the second pulse **338b** of the TURN-ON transition **304** is long enough to cause filter/delay circuit **606** to generate a pulse **716** for the TDLY_FLTR signal **609**. With both the TDLY_FLTR signal **609** and the output signal of AND gate **608** being logic high, the SET signal **613** becomes logic high at rising edge **720**. With the S terminal of SR flip-flop **619** being logic high, SR flip-flop **616** causes the PS_ON_PRI signal to transition to a logic high state at rising edge **345**, thereby causing control and fault management **208** to cause power stage **206** to turn on power transfer through isolator **120**.

[0052] The rising edge **722** of the TDLY_FLTR signal **609** causes D flip-flop **622** to clock the logic 1 at its D input terminal through to its Q output terminal. Delay circuit **624** delays the rising edge at the signal at its input terminal **624a** by, for example, 750 ns, to produce a pulse **742** for the TDLY2 signal **625**. PG **626** responds to the rising edge **744** of pulse **742** at its clock terminal by generating a pulse **312** (e.g., 25 ns wide) for FAULT_DECODER signal **311** into the clock terminal of D flip-flop **628**. When clocked, D flip-flop **628** generates the FAULT_STATE signal **245** to have the same logic state as the logic state of the RX_OUT_PLS signal **237** when the rising edge **748** of pulse **312** occurs. In the example of FIG. **7**, when rising edge **748** occurs, the logic state of the RX_OUT_PLS signal **237** is logic low as a result of falling edge **665** which corresponds to falling edge **349** of the TX_ON_PLS signal **233** in FIG. **3**. Accordingly, in this example, the logic state of the FAULT_STATE signal **245** is a logic 0, which indicates, as described above, that power is being transferred from first circuit **110** to second circuit **130** through isolator **120**.

[0053] If switching detect circuit **226** had not detected the occurrence of power transfer through isolator **120** into second circuit **120**, then switching detect circuit **226** would not have asserted the SEC_SW_DET signal **235** to a logic high state and PG **403** (FIG. **4**) would not have been reset. If PG **403** is not reset by a logic high state for the SEC_SW_DET signal **235**, then the second pulse **304b** of the TURN-ON transition **304** in FIG. **3** will not have a falling edge until falling edge **669** at the end of the full time period of the pulse generated by PG **403** (e.g., a pulse of 1 μ s). In that case, the logic state of the FAULT_STATE signal **245** will be a logic 1 because the RX_OUT_PLS signal **237** will still be at a logic high state when D flip-flop is clocked by the rising edge **748** of the FAULT_DECODER signal **311**.

[0054] FIG. **8** is a block diagram of isolated power converter **100** in another example capable of generating a multi-bit fault indication and transmitting the multi-bit fault indication from second circuit **130** to first circuit **110**. The isolated power converter **100** of FIG. **8** is largely the same as that of the isolated power converter of FIG. **2**, and its description is not repeated here. The isolated power converter **100** of FIG. **8** also includes a fault controller **810** that provides an output FAULT_SEC signal **812** to controller **138**. In the example of FIG. **8**, a temperature sensor **825** is included and coupled to fault controller **810**. The output voltage VOUT is also provided to fault controller **810**. Fault controller **810** generates a multi-bit fault indication to represent any of multiple fault conditions. Examples of such fault conditions include (a) whether second circuit **130** is receiving power from first circuit **110** (as described above), (b) an over-temperature fault, (c) whether output voltage VOUT is too low (undervoltage fault), etc. In one example, fault controller

810 receives input signals representing fault states and generates the multi-bit fault indication. The multi-bit fault indication may include any number of bits, e.g., three bits. A 3-bit fault indication can represent eight different fault conditions. Fault controller **810** controls the logic state of the FAULT_SEC signal **812** to cause controller **138** to generate each bit of the multi-bit fault indication. Controller **138** encodes each bit of the multi-bit fault indication in the logic state at the end of pulse **304b**, as described above. Controller **138** generates multiple sets of the TURN_ON indication **304** and the TURN_OFF indication **354** to transmit the multi-bit fault indication. Each TURN_ON indication **304** includes a fault bit of the multi-bit fault indication.

[0055] Fault controller **810** prepends the multi-bit fault indication with, e.g., a logic “1” as a flag to indicate to control and fault management **208** that the subsequent, e.g., three bits represent the fault indication. The flag bit and the bits of the multi-bit fault indication are transmitted serially through coils L4 and L3 of isolator **120** from the modulator **222** to the demodulator **202** as part of each subsequent TURN_ON indication **304**. Control and fault management **208** detects the presence of the flag bit and then receives the subsequent bits forming the multi-bit fault indication, FAULT_STATE<n:0> **245**. Control and fault management **208** may respond to the fault indication in any suitable manner, e.g., by turning off power stage **206**. Upon receiving the full multi-bit fault indication FAULT_STATE<n:0> **245**, control and fault management **208** generates a FAULT_RESET signal **831** to decoder **118**.

[0056] FIG. **9** is a circuit schematic of controller **138** suitable for use in generating a multi-bit fault indication. Controller **138** in FIG. **9** is largely the same as controller **138** in FIG. **4** and its full description is not repeated here. Controller **138** in FIG. **9** also includes an AND gate **910** and a NOR gate **912**. One input of each of AND gate **910** and NOR gate **912** receives the SEC_SW_DET signal **235**. The FAULT_SEC signal **812** is provided to an inverted input of AND gate **910** and to the other input of NOR gate **912**. The output of AND gate **910** is coupled to the reset terminal of PG **403**. The output of NOR gate **912** is coupled to the reset terminal of PG **404**.

[0057] PG **403** is reset thereby forcing the PLS3 signal **414** to a logic low if the FAULT_SEC signal **812** is logic low and the SEC_SW_DET signal **235** is logic high. Otherwise, PG **403** will force the PLS3 signal **414** logic low at the end of its time period (e.g., 1 μ s). If either of the SEC_SW_DET signal **235** or the FAULT_SEC signal **812** are logic high, NOR gate **912** generates a logic 0 for its output signal to the reset terminal of PG **404**.

[0058] FIG. **10** is a circuit schematic of decoder **118** suitable for use in decoding a multi-bit fault indication. Decoder **118** in FIG. **10** is largely the same as decoder **118** in FIG. **6** and its full description is not repeated here. Decoder **10** in FIG. **10** also includes additional D flip-flops **1002**, **1004**, and **1006**. The combination of D flip-flops **628**, **1002**, **1004**, and **1006** form a shift-register to shift in each bit of the multi-bit fault indication. The output signal from D flip-flops **628**, **1002**, **1004**, and **1006** are bits 0, 1, 2, and 3, respectively, of the multi-bit fault indication (assuming a 3-bit fault indication). Upon receipt of all of the bits of the multi-bit fault indication, control and fault management **208** asserts (e.g., logic high) the FAULT_RESET signal **831** to reset D flip-flops **628**, **1002**, **1004**, and **1006** to prepare them to receive the next multi-bit fault indication.

[0059] FIG. **11** is a timing diagram in another example in which a multi-bit fault indication can be transmitted through isolator **120** and decoded by decoder **118**. Each bit of the multi-bit fault indication is recovered by decoder **118** in a separate cycle of the TURN-ON and TURN-OFF indications. The example timing diagram of FIG. **11** includes the same signals as in the timing diagram of FIG. **3**. Each instance of a pulse **312**, shown as pulses **312a-312d** in FIG. **8**, of the FAULT_DECODER signal **311** causes D flip-flop **628** to latch the logic state of the RX_OUT_PLS signal **237** as the next bit of the multi-bit fault indication. Each bit latched by D flip-flop **628** is sequenced through the other D flip-flops **1002**, **1004**, and **1106** upon each pulse of the FAULT_DECODER signal **311**. In the example of FIG. **11**, the FAULT_STATE signal **245** has a 4-bit value with each of the four consecutive pulses **312a-312d** causing one of the four bits of the 4-bit value to be recovered. The first bit (e.g., most significant bit) may be the flag bit described

above indicating the beginning of a multi-bit fault indication. The first pulse **312a** results in bit **801** of the FAULT_STATE signal **245** (e.g., the flag bit). The second pulse **312b** results in bit **802** of the FAULT_STATE signal **245**. The third and fourth pulses **312c** and **312d** result in bits **803** and **804**, respectively, of the FAULT_STATE signal **245**. Bits **802**, **803**, and **804** represent the fault (e.g., power not being transferred through isolator **120**, over-temperature, undervoltage, etc., or combinations thereof). In the example of FIG. **11**, each of the bits is a logic “1” because fault controller **810** has held the FAULT_SEC signal **812** logic high thereby preventing PG **403** from being reset and allowing the PLS3 signal **414** to be extended to the end of the time period of PG **403**. If fault controller **810** had forced the FAULT_SEC signal **812** logic low for any of the bits, then the logic state of the SEC_SW_DET signal **235** would have dictated the logic state of the corresponding fault bit.

[0060] FIG. **12** is a schematic diagram of switching detect circuit **226**, in an example. FIG. **13** is a timing diagram illustrating the operation of switching detect circuit **226**. In this example, switching detect circuit **226** includes comparators **1201** and **1202** and an OR Gate **1203**. The positive input of comparator **1201** and the negative input of comparator **1202** are coupled to terminals **125** and **126**, respectively, of isolator **120**. The negative input of comparator **1201** and the positive input of comparator **1202** are coupled to terminals **126** and **125**, respectively, of isolator **120**. The voltage on terminal **125** is SW_POS and the voltage on terminal **126** is SW_NEG. Each comparator **1201** and **1202** compares SW_POS to SW_NEG. Voltage V.sub.B represents a fixed offset voltage so that each comparator outputs a logic high signal if the signal at the positive terminal is larger than the SW_NEG or SW_POS signal provided to the negative terminal by at least the voltage V.sub.B. Comparator **1201** outputs a logic high if SW_POS is larger than SW_NEG by voltage V.sub.B. Comparator **1202** outputs a logic high if SW_NEG is larger than SW_POS by voltage V.sub.b.

[0061] FIG. **13** illustrates switching across coil L2 when power is being transferred through isolator **120**. OR gate logically ORs the output signals from comparators **1201** and **1202**. During each cycle of the switching waveform of FIG. **13**, one or the other of comparators **1201** and **1202** outputs a logic high signal and thus the output signal from OR gate **1203**, SEC_SW_DET **235**, is logic high during power transfer through isolator **120**.

[0062] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0063] As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0064] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0065] Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other

form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0066] Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. A power converter, comprising: an isolator having a first terminal, a second terminal, a third terminal, and a fourth terminal; a first circuit having a first terminal coupled to the first terminal of the isolator, and having a second terminal coupled to the second terminal of the isolator; and a second circuit having a first terminal coupled to the third terminal of the isolator, and having a second terminal coupled to the fourth terminal of the isolator, the second circuit including a controller configured to transmit a control signal through the isolator to the first circuit, the control signal including a first indication to turn on power transmission through the isolator from the first circuit to the second circuit, a second indication of a status, and a third indication to turn off power transmission.
2. The power converter of claim 1, wherein: the first indication includes a first signal pulse followed by a second signal pulse, the second signal pulse is longer than the first signal pulse.
3. The power converter of claim 2, wherein the third indication includes a single pulse.
4. The power converter of claim 2, wherein the status is indicated in a length of the second signal pulse.
5. The power converter of claim 2, wherein the controller determines that the second circuit is receiving power and, in response, provides the second indication by terminating the second pulse.
6. The power converter of claim 1, wherein the status indicates whether the second circuit is receiving power.
7. The power converter of claim 1, wherein: the status of the second indication is a status bit; the control signal includes multiple sets of the first indication, the second indication, and the third indication; and the controller is configured to transmit a multi-bit fault indication through the isolator, each bit of the multi-bit fault indication is the status bit of the second indication of a separate set.
8. The power converter of claim 7, wherein the multi-bit fault indication indicates at least one of whether the second circuit is receiving power, an over-temperature fault, or an undervoltage fault.
9. The power converter of claim 1, wherein the controller includes: a first pulse generator having an output; a second pulse generator having an input coupled to the output of the first pulse generator, and having an output; and a third pulse generator having an input coupled to the output of the second pulse generator.
10. The power converter of claim 9, wherein the third pulse generator is configured to generate a longer pulse than the second pulse generator, and the second pulse generator is configured to generate a longer pulse than the first pulse generator.
11. A power converter, comprising: an isolator having a first terminal, a second terminal, a third terminal, and a fourth terminal; a first circuit having a first terminal coupled to the first terminal of the isolator, and having a second terminal coupled to the second terminal of the isolator; and a second circuit having a first terminal coupled to the third terminal of the isolator, and having a second terminal coupled to the fourth terminal of the isolator; wherein the first circuit includes a decoder configured to decode a control signal received from the second circuit through the isolator, the control signal including a first indication to turn on power transmission through the isolator to the second circuit, a second indication of a status of the second circuit, and a third indication to turn off power transmission.

- 12.** The power converter of claim 11, wherein the first indication includes a first signal pulse followed by a second signal pulse, the second signal pulse is longer than the first signal pulse, and the third indication includes a single pulse.
- 13.** The power converter of claim 12, wherein the status is indicated in a length of the second signal pulse.
- 14.** The power converter of claim 11, wherein the status indicates whether the second circuit is receiving power.
- 15.** The power converter of claim 11, wherein: the second indication is a status bit; the control signal includes multiple sets of first, second and third indications; and the controller is configured to determine a multi-bit fault indication, each bit of the multi-bit fault indication is the status bit of the second indication of a separate set.
- 16.** The power converter of claim 15, wherein the multi-bit fault indication indicates at least one of whether the second circuit is receiving power, an over-temperature fault, or an undervoltage fault.
- 17.** The power converter of claim 15, wherein the decoder includes: a delay having an input configured to receive the control signal, and having an output; and a flip-flop having a clock input coupled to the output of the delay, and having a data input configured to receive the control signal.
- 18.** A power converter, comprising: an isolator having a first terminal, a second terminal, a third terminal, and a fourth terminal; a first circuit having a first terminal coupled to the first terminal of the isolator, and having a second terminal coupled to the second terminal of the isolator; and a second circuit having a first terminal coupled to the third terminal of the isolator, and having a second terminal coupled to the fourth terminal of the isolator, the second circuit including a controller configured to transmit a control signal through the isolator to the first circuit, the control signal including a first indication to turn on power transmission through the isolator from the first circuit to the second circuit, a second indication of whether the second circuit is receiving power from the isolator, and a third indication to turn off power transmission, the second indication indicated in a length of a signal pulse; wherein the first circuit receives the control signal from the second circuit through the isolator and determines the second indication.
- 19.** The power converter of claim 18, wherein: the control signal includes multiple sets of first, second and third indications; and the first circuit is configured to determine a multi-bit fault indication, each bit of the multi-bit fault indication included in a second indication of a separate set.
- 20.** The power converter of claim 18, wherein the first indication includes a first signal pulse followed by a second signal pulse, the second signal pulse is longer than the first signal pulse, and the third indication includes a single pulse.
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